Kuanwei Chen

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Summary

M.S. Computer Science student at National Central University specializing in Artificial Intelligence, with research presented at three international conferences. Experienced in building computer vision and multimodal deep learning systems for scene text recognition, sign language translation, and biometric recognition using PyTorch and TensorFlow. Built an open-source sign language data pipeline with 100+ GitHub stars, and seeking AI/ML engineering or computer vision research roles.

Education

National Central University, Taoyuan, Taiwan *Starting Sep 2025*
Master in Computer Science, specialization in Artificial Intelligence

National Changhua University of Education, Changhua, Taiwan *Sep 2021 – Jun 2025*
Bachelor in Electrical Engineering, GPA 3.8/4.0 | Grad. Cert.

Experience

Project Assistant in AI Application Laboratory *Jun 2024 – Dec 2024*
Certificate

- Collaborated with 8 international interns on multimodal AI projects spanning biometric recognition and action recognition using PyTorch and TensorFlow frameworks under the TEEP and IIPP research programs, delivering functional prototypes that were showcased to faculty and contributed to a conference submission.
- Built Pandas and NumPy-based preprocessing, dataset, and evaluation pipelines that streamlined model experimentation and validation across computer vision and speech-related research tasks.

Research Publication

STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition *IEEE ICCE-TW*
ResearchGate | GitHub

- Co-first author; designed a 6M-parameter Vision Transformer (ViT) with Masked Autoencoder (MAE) pretraining and fine-tuning with an autoregressive decoder for Scene Text Recognition.
- Achieved 93.82% weighted accuracy on six standard Computer Vision benchmarks and 81.03% on U14M.

Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs *IEEE ICCE-TW*
arXiv | ResearchGate

- First author; built a 77M-parameter end-to-end Natural Language Processing (NLP) pipeline.
- reduce computation by ~75% with minimal BLEU impact (10.06→9.53), providing efficiency trade-off insights.

Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition *IET ICETA*
Publication | arXiv | ResearchGate | GitHub | Certificate

- First author; led a team of 2 interns in designing a multi-modal AI/ML system combining Machine Vision and Audio Recognition/Speaker Verification.
- Designed a ResNet-based Convolutional Neural Network (CNN) with Fbank feature extraction and triplet loss for voice verification (99.1%); fine-tuned VGG16 with MTCNN object detection in TensorFlow (95.1%).

Projects

SignDATA: Data Pipeline for Sign Language Translation *Python, Det, Pose, System Design*
arXiv | ResearchGate | GitHub

- First author; open-source repo with 100+ stars; built a config-driven data preprocessing, and dataset management system for standardizing heterogeneous sign-language corpora for AI/ML Pipeline development.
- Designed processing pipeline integrating YOLO/MMDet detection and MediaPipe/MMPose landmark extraction for structured feature engineering, applying system-design principles for multi-dataset scalability.

Skills

Spoken Languages	English (IELTS 6.5, TOEIC 830), Mandarin (native), Hokkien (native)
Programming & Tools	Python, C/C++, HTML, CSS, JavaScript, Git, GitHub, React, Vue
Frameworks & Libraries	PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas
AI/ML	Machine Learning, Deep Learning, Vision-language Models, Multimodality, AI/ML Pipeline Development, Data Preprocessing, Dataset Management, Feature Engineering, Transfer Learning, Fine-tuning, System Design
Computer Vision & NLP	Computer Vision, Natural Language Processing (NLP), Biomedical Engineering, Action Recognition, Speech Recognition, Object Detection
Models & CV Stack	Convolutional Neural Network, Transformer, YOLO, MMDet, MediaPipe